

Homework 2

1. Method Description

1.1. Overview of PrivBayes

PrivBayes by Zhang et al., 2017 is a differentially private data release mechanism designed for tabular datasets. It has two stages, it learns a Bayesian network to capture the joint distribution of the dataset's attributes while adding calibrated Laplace noise to satisfy DP. Then it uses a privatized network to generate a fully synthetic dataset that statistically resembles the original data.

1.2. Bayesian Network Structure

PrivBayes models the dataset as a Bayesian Network where each attribute is a node and each has directed edges to represent the conditional dependencies. It uses greedy algorithm to select parent nodes for each attribute using the exponential mechanism, ensuring that the structure selection itself is differentially private. In this experiment, the maximum number of parent nodes k was set to 2 to balance expressiveness with computational efficiency.

2. Dataset and Preprocessing

2.1. Dataset

The dataset that was used for this experiment is the same with the dataset that was used in homework 1, which is The Spaceship Titanic dataset from Kaggle. The dataset contains 8,693 passenger records from a fictional interstellar vessel. The classification task whether a passenger was transported to an alternate dimension. The dataset contains numerical, categorical, and complex structured column

2.2. Preprocessing

- Dropped all rows with null values
- Removed duplicate rows
- Parsed the Cabin column (format: 'Deck/CabinNum/Side') into three separate columns
- Parsed the PassengerID column (format: 'GroupId_MemberId') into two separate columns
- Dropped the Name column due to its high cardinality (almost each row has unique values)

- Encoded the boolean columns to an integer (0 and 1)
- After synthesis, applied One-Hot Encoding to categorical columns: HomePlanet, Destination, Deck, and Side

3. Experimental Results

3.1. Results

Epsilon	Accuracy (%)	Precision (%)	Recall (%)	AUC (%)
Baseline	81.85	82.35	81.61	81.85
0.1	51.36	51	98.95	50.78
0.5	49.17	48.97	10.61	49.64
1.0	50.45	50.53	99.25	49.86
5.0	72.24	82.13	57.7	72.42
10.0	72.24	81.23	57.7	72.42

3.2. Results

As expected from differential privacy theory, lower epsilon produces more private but less accurate synthetic data, while higher epsilon values produce data that more closely approximates the utility of the real dataset. At epsilon = 0.1, the strong noise injection degrades the learned distribution significantly leading to lower classification performance. As epsilon increases toward 10.0, the synthetic data increasingly resembles the real data in its statistical structure.

4. Novel Ideas and Observations

4.1. Post-Synthesis One Hot Encoding

A key design decision in this project was to apply One-Hot Encoding after the synthesis step rather than before. In HW1, OHE was applied directly to the raw dataset before any privacy transformation. However, when OHE is applied before feeding data into PrivBayes, the Cabin column alone (with hundreds of unique CabinNum values) would produce hundreds of sparse binary columns, making the Bayesian network computationally intractable and statistically meaningless. By feeding categorical strings directly into PrivBayes and applying OHE afterward,

we preserve the semantic structure of the data and dramatically reduce the computational burden.

4.2. Library Bug Fix

During implementation, a bug was discovered in the DataSynthesizer library (DataGenerator.py) where the eval() function failed on single-parent Bayesian network nodes, raising a NameError. The root cause was that eval() on a single-value string returns an integer rather than a tuple, making list() conversion fail. A targeted patch was applied to handle both single-parent and multi-parent cases, enabling successful synthetic data generation.

5. Comparison with HW1

5.1. Privacy Mechanism

Aspect	HW1	HW2
Privacy Model	Syntactic Privacy	Formal, DP guarantee
Output Type	Generalized real records	Fully synthetic dataset
Data Distribution	Suppression and generalization of values	Statistical noise via Laplace mechanism
Attacker Model	Assumes attacker knows quasi identifiers	Bounds any attack with arbitrary background knowledge
Composability	Not composable	Fully composable with privacy budget
Utility Impact	Removes or blurs specific records	Trades off utility via epsilon tuning

5.2. Data Utility

In HW1, k-anonymity and l-diversity were applied to protect quasi-identifiers by suppressing or generalizing records that could not form groups of size at least k. This approach preserves utility for the remaining records but removes data points entirely. In HW2, PrivBayes generates a completely synthetic dataset of the same size as the original, meaning no real records are retained at all, instead all data is

generated from a privatized statistical model. At higher epsilon values, the synthetic data tends to achieve utility comparable to or exceeding the HW1 obfuscated dataset, while also providing stronger formal privacy guarantees.

6. Comparison with Midterm

The midterm implemented Deep Learning with Differential Privacy using DP-SGD (Differentially Private Stochastic Gradient Descent), as described in Abadi et al. (2016). In DP-SGD, differential privacy is enforced during the model training process itself: per-sample gradients are clipped to a maximum L2 norm, and Gaussian noise is then added to the aggregated gradients at each training step. This ensures that the trained model does not memorize or leak information about any individual training record.

In contrast, HW2 applies differential privacy at the data level using PrivBayes, before any model training occurs. The two approaches represent fundamentally different points of DP injection in the machine learning pipeline:

Aspect	Midterm	HW2
Where DP is applied	During model training	During data generation
Mechanism	Gradient clipping + Gaussian noise	Exponential mechanism + Laplace noise
Privacy guarantee	The trained model is private	The released dataset is private
Real data used?	Yes	No
Output	A privacy-preserving trained model	A privacy-preserving dataset
Downstream flexibility	One specific model trained privately	Synthetic data can be shared and reused freely
Privacy parameter	Epsilon computed via moments accountant	Epsilon set directly as input to PrivBayes
Noise target	Gradients	Conditional probability distributions

The key philosophical difference is that DP-SGD protects privacy by ensuring the model cannot be reverse-engineered to reveal training data, while PrivBayes protects privacy by ensuring the original data is never directly shared at all. DP-SGD is more appropriate when the goal is to publish a trained model; PrivBayes is more appropriate when the goal is to share a dataset for general-purpose downstream use. Both approaches provide rigorous epsilon-differential privacy guarantees, but they operate at different stages and serve different use cases in the data science pipeline.